Power Quality Disturbance Classification using Machine Learning

Presented by

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1. **Project Area** : Power Systems & Machine Learning
2. **Project Title** : Power Quality Disturbance Classification using Machine Learning
3. **Place of Work** : In-house
4. **Supervisor Name** : Mr. N. Suresh
5. **Project Group** : Batch 15

Motivation

- There is a clear gap in high accuracy ML models available which has low feasibility.
- **Industries need low-latency, edge-compatible intelligence**, with increasing renewable integration, **grids are becoming more unstable**, making real-time PQ monitoring essential.
- **Traditional techniques (FFT, RMS)** fail to capture complex, transient, and non-linear disturbances
- A scalable solution must **balance accuracy, speed, and computational efficiency -** current systems don't.
- This project targets a practical shift, from delayed detection to real-time intelligent classification at the edge.

Problems Identified and Solutions Suggested

Problems Identified

- Frequent PQ Disturbances: Over **60%** of grid issues stem from voltage sags, swells & harmonics causing **15–20% energy loss** and **30% shorter equipment life**.
- Outdated Detection: Conventional FFT/RMS systems **miss ~40% of transients** and analyze events **5–10 seconds late**, risking downtime.
- High Computational Demand: Existing AI models need **>2 GB GPU RAM** and cloud connectivity, adding **300–400 ms latency** impractical for field deployment.

Solutions Suggested

- Accuracy: Classifies PQ disturbances with **>95% accuracy** and **<100 ms inference time.**
- Edge AI Implementation:Cutting latency by **80%** and power cost by **40%**.
- IoT-Based Monitoring: Sensors stream **10 kHz data** to a live Node-RED dashboard, improving response **speed by ~70%**.

Literature Survey

PAPER TITLE & AUTHOR	WORK	OUTCOME
Power Quality Disturbance Classification using Random Forest with Feature Selection for Real-Time Monitoring – Rahul, A. Sambher & P. Sinha, Lecture Notes in Electrical Engineering, Springer, 2025	Applied Random Forest classifier on PQD dataset with RMS, THD, crest factor, and spectral band features; studied feature count vs. accuracy trade-off for real-time deployment.	Achieved 97.50% accuracy — confirms RF is effective for real-time PQ classification but tested only on single disturbances without compound classes like Sag+Harmonics.
Power Quality Disturbances Categorization using Identity Feature Vector and Extreme Learning Machine – H. Rezapour et al., Results in Engineering, Elsevier, 2024	Generated 12 PQD classes including Sag, Swell, Harmonics, Flicker, Sag+Harmonics, Sag+Flicker, and Swell+Harmonics using wavelet-based identity feature vectors classified by ELM.	10 out of 12 classes achieved 100% accuracy; Swell+Harmonics was the hardest at 98% — our work addresses this using Random Forest with stronger spectral features.
Detection and Classification of PQDs using SPWVD and Vision Transformer – K. Ozbay & M. Gol, AIMS Energy, vol. 13, no. 5, 2025	Simulated 11 PQD types per IEEE 1159 in MATLAB (sag, swell, interruption, harmonics, flicker, harmonics with sag/swell, flicker with sag/swell) and classified with ViT model.	99.86% overall accuracy but requires image transformation and heavy GPU — our RF approach works directly on extracted features with no image conversion.

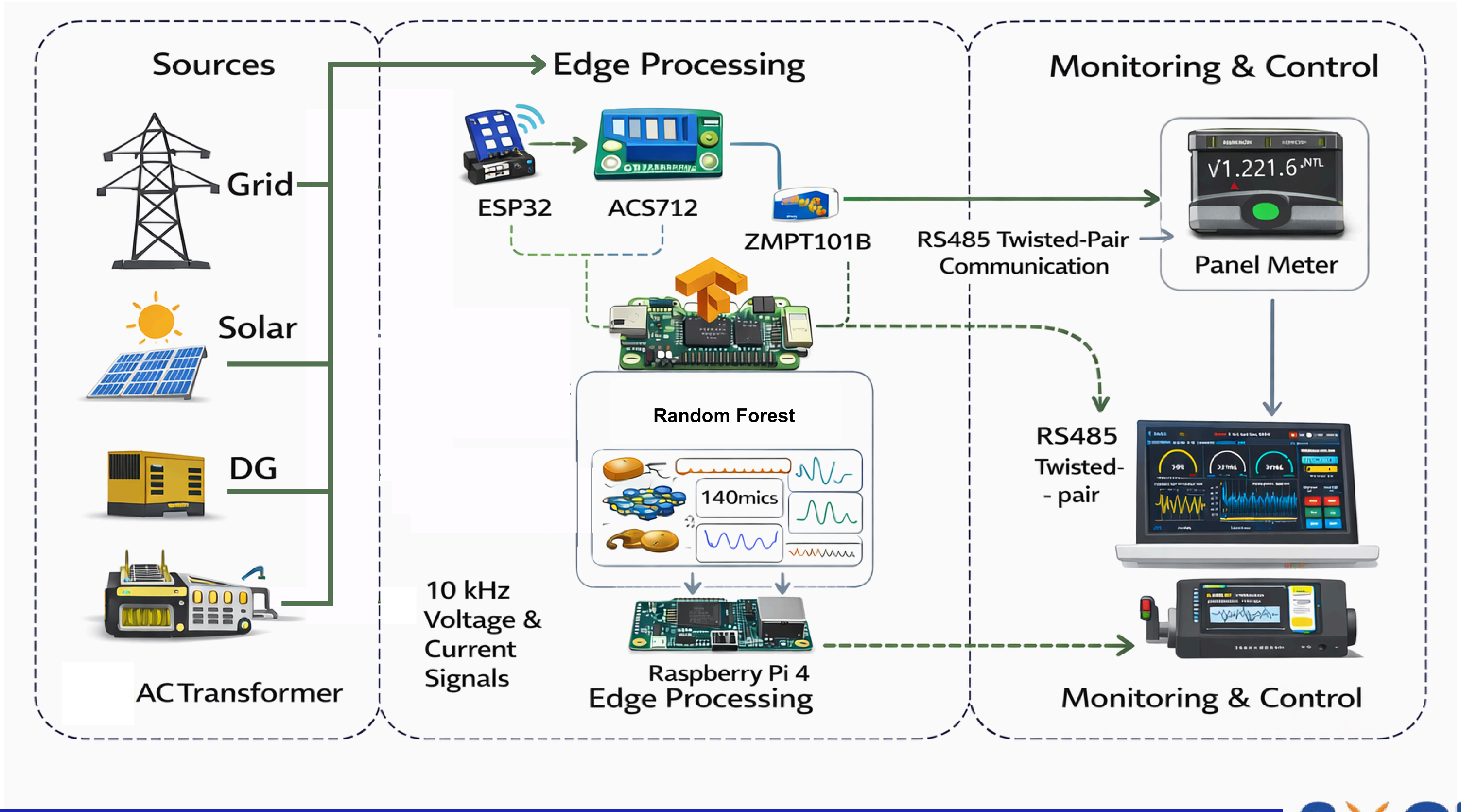
Literature Survey

PAPER TITLE & AUTHOR	WORK	OUTCOME
A Comprehensive Research of Machine Learning Algorithms for PQD Classifier Based on Time-Series Window – S. S. Tezcan et al., Electrical Engineering, Springer, 2025	Benchmarked RF, XGBoost, LightGBM, and kNN on a 21-class PQD dataset including compound disturbances at SNR 20–50 dB per IEEE 1159 and IEC 61000 standards.	RF delivered competitive accuracy — validates our choice of RF as classifier for a 12-class compound PQD problem.
Perception of PQDs using Fourier, STFT, CWT and DWT – S. Lather et al., Scientific Reports, Nature, Feb. 2024	Mathematically modelled Sag, Swell, Interruption, Harmonics, Flicker in MATLAB using parametric equations per IEEE 1159; analysed signals using FFT, STFT, CWT, and DWT transforms.	Validated MATLAB signal generation equations for all PQD types used in our project; confirmed FFT-based features (THD, harmonic magnitudes) are most discriminative.

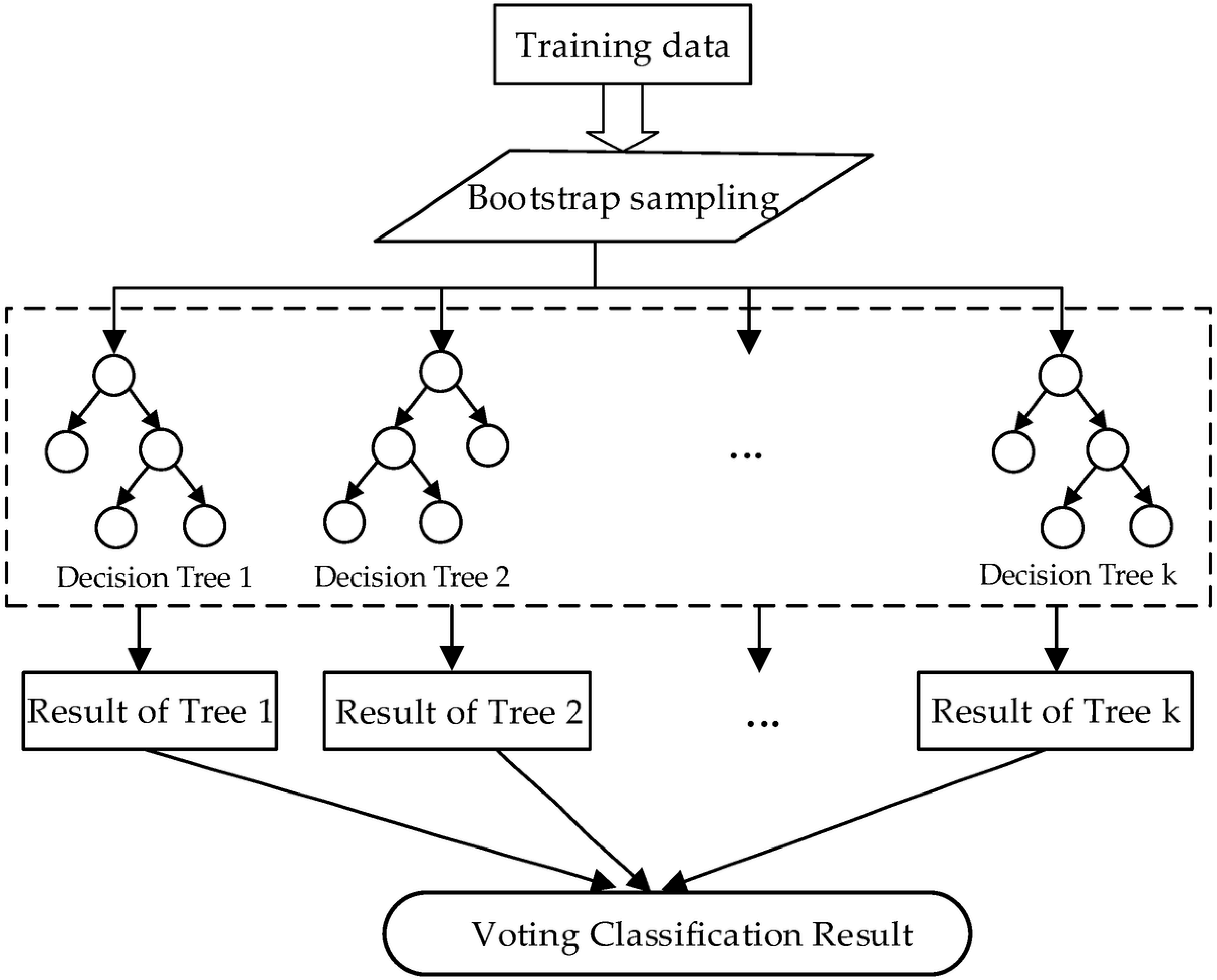
Objectives of the Project

- Develop a robust machine learning model to accurately **classify multiple types of power quality disturbances** under real-world conditions.
- **Achieve high classification accuracy (>95%) with low inference time (<100 ms)** for practical deployment.
- **Enable edge-compatible implementation**, reducing dependency on high-end GPUs and cloud infrastructure.
- Extract and **optimize multi-domain features (time, frequency, wavelet)** to improve model reliability across disturbance types.
- Provide clear output including disturbance type, severity indication, and confidence score for actionable insights.

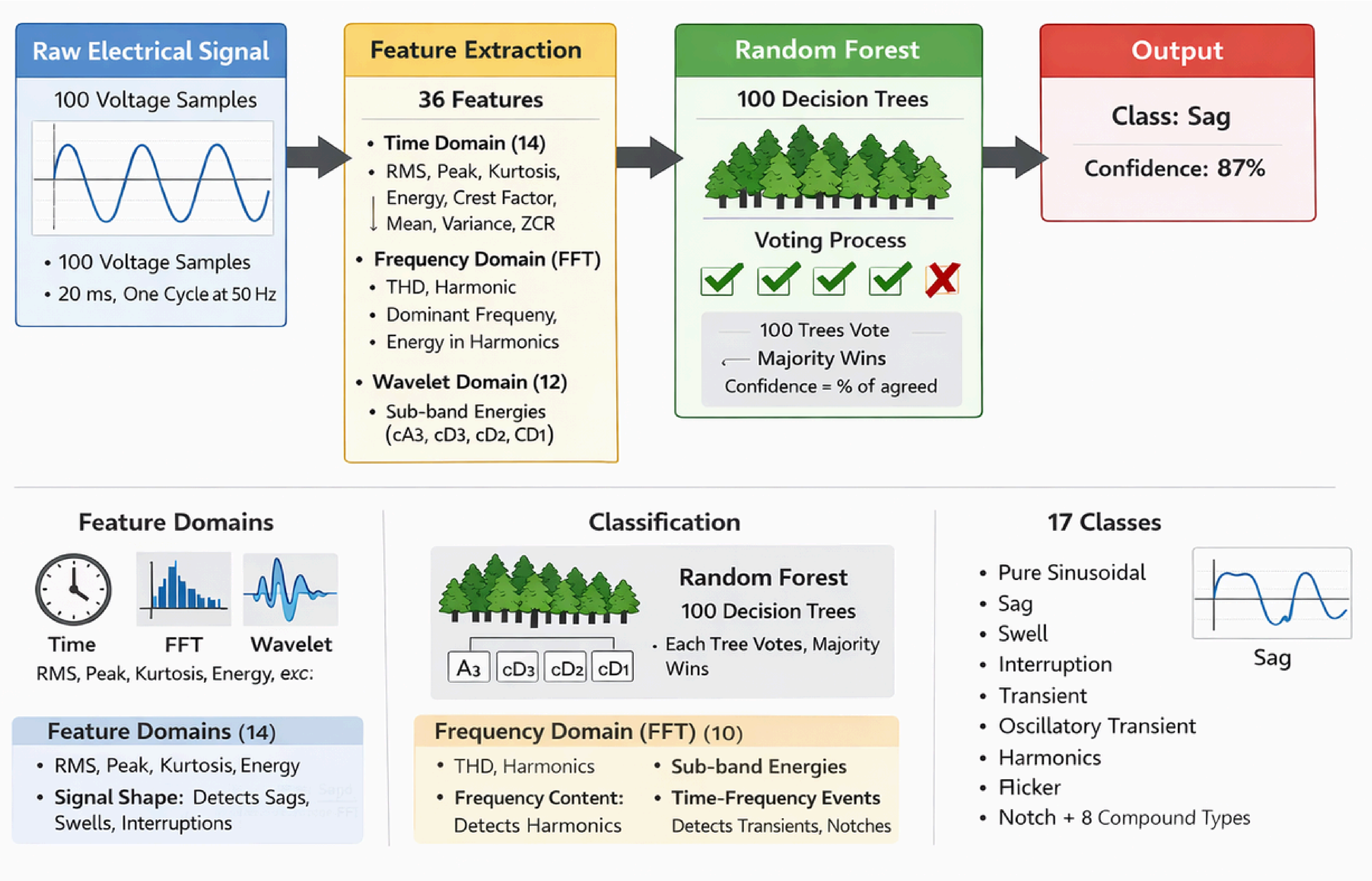
Architecture



Random Forest



System Design



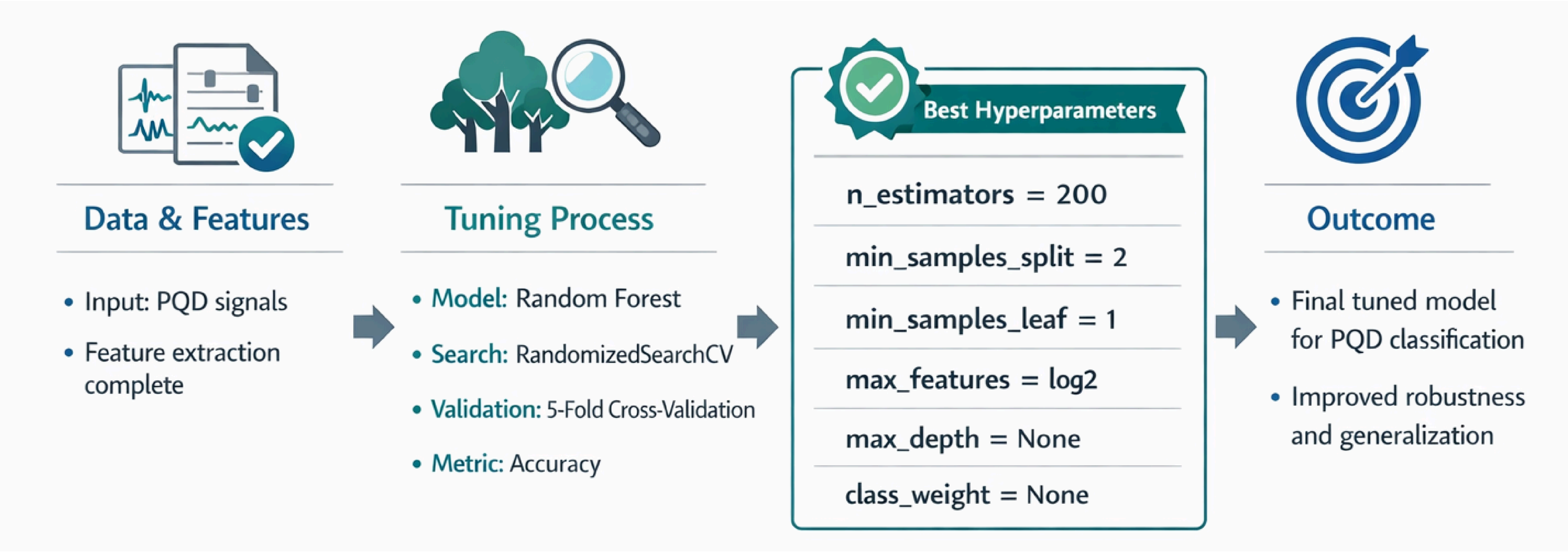
Training Phase (done once):

- Load XPQRS dataset (17,000 signals, 17 classes, 100 samples each)
- Extract 36 features from each signal (time + FFT + wavelet)
- Split 80% train / 20% test
- Normalize features using StandardScaler
- Train Random Forest classifier (100 decision trees)
- Validate with 5-fold cross-validation
- Save trained model to disk (.pkl)

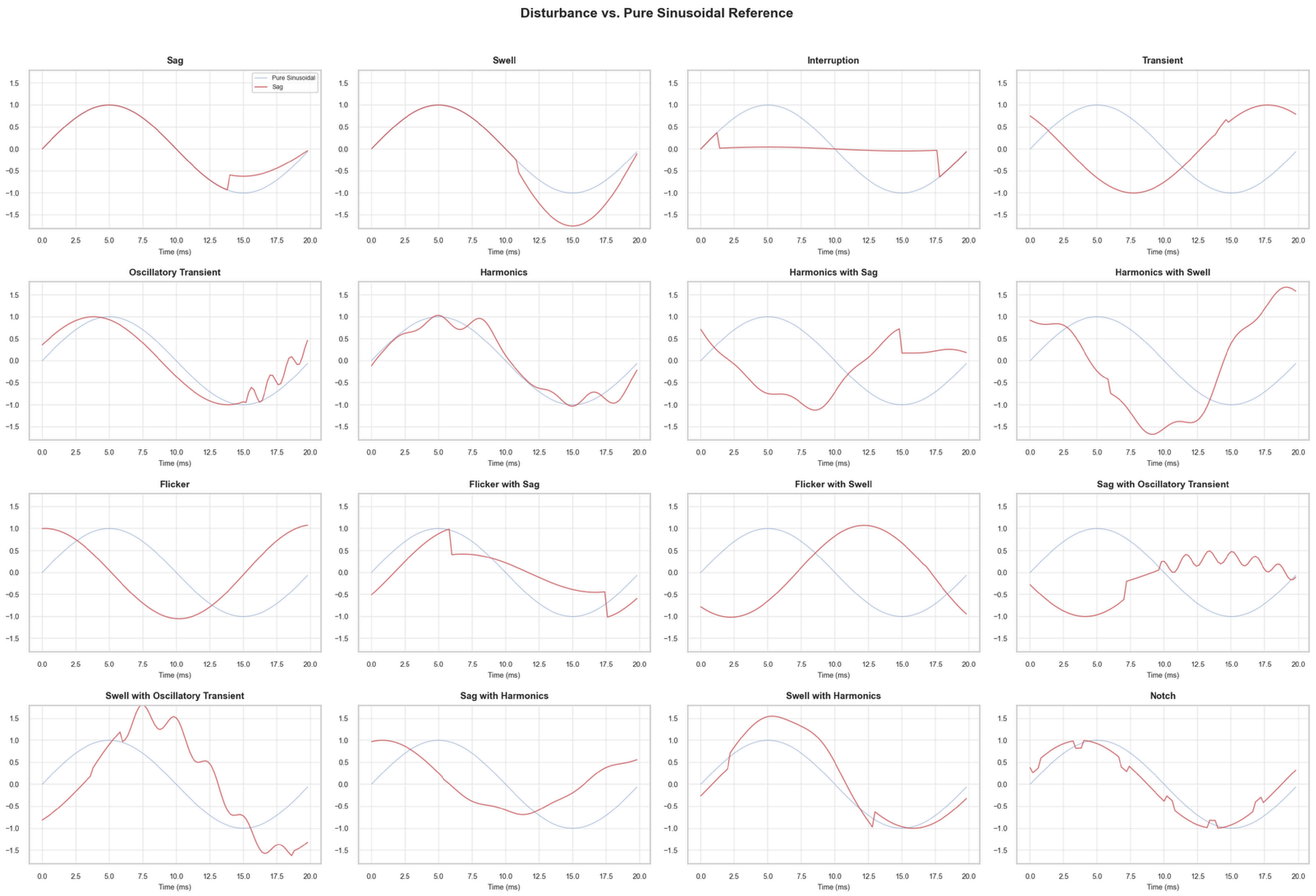
Prediction Phase (runs anytime):

- Take new signal (100 voltage values)
- Extract 36 features
- Load saved model and predict
- Output: Normal/Abnormal + disturbance type + confidence score

Hyperparameter Tuning



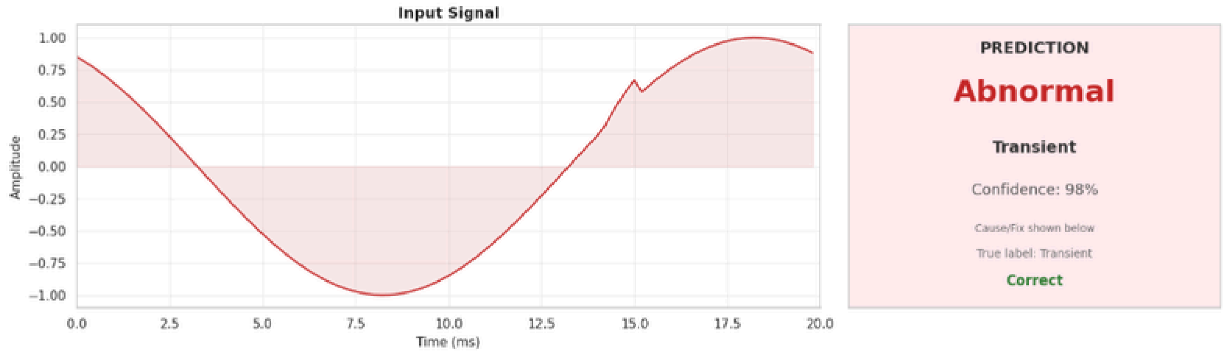
Disturbances



Output

Transient (sharp spike)

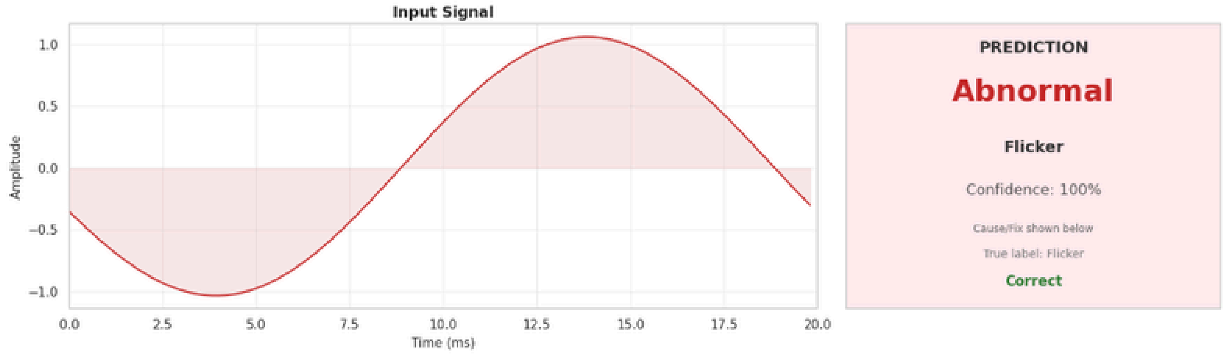
```
[7]: idx = (labels == 'Transient').nonzero()[0][15]
     _ = demo_predict(signals[idx], 'Transient')
```




```
Likely cause: Fast high-amplitude spike from lightning, switching events, or fault clearing.
Recommended fix: Install surge protection devices, improve grounding, and review switching practices.
```

Flicker (voltage fluctuation)

```
[8]: idx = (labels == 'Flicker').nonzero()[0][8]
     _ = demo_predict(signals[idx], 'Flicker')
```




```
Likely cause: Low-frequency voltage fluctuation from rapidly varying loads like welders or arc furnaces.
Recommended fix: Use STATCOM/SVC or dynamic voltage support and separate highly fluctuating loads.
```

Plan of Action

	DECEMBER 2025	JANUARY 2026	FEBRUARY 2026	MARCH 2026
TASK				
Literature Survey				
Design				
Simulation				
Simulation and Software Integration				
Report Preparation				

SDG

SDG Goals	Project Contribution
SDG 7 – Affordable and Clean Energy	Improves energy efficiency and grid stability by enabling real-time detection of power quality disturbances, ensuring reliable and sustainable electricity supply.

References

[1] A.Ali Memon, M. Ali Koondhar, Saad F. Al-Gahtani, Z.M.S. Elbarbary and Z.Muhammed Alaas, "Comprehensive review of power quality disturbance detection and classification techniques,"Computers and Electrical Engineering,Science Direct, volume 126, August 2025.

[2] S. Janthong and P. Phukpattaranont, "Power quality disturbance detection using improved grasshopper optimization and adaptive boosted random forest," Electrical Engineering, Springer, 2025.

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[6] R. J. J. Molu, W. F. Mbasso, K. T. Saatong, and S. R. D. Naoussi, "Enhancing power quality monitoring with discrete wavelet transform and extreme learning machine: a dual-stage pattern recognition approach," Frontiers in Energy Research, vol. 12, p. 1435704, 2024.

[7] S. Jain, A. Satsangi, R. Kumar, and D. Panwar, "Intelligent assessment of power quality disturbances: a comprehensive review on machine learning and deep learning solutions," Computers and Electrical Engineering, vol. 123, Elsevier, 2025.

[8] M. S. Priyadarshini, M. Bajaj, L. Prokop, and M. Berhanu, "Perception of power quality disturbances using Fourier, short-time Fourier, continuous and discrete wavelet transforms," Scientific Reports, Nature, vol. 14, p. 3443, Feb. 2024.

[9] M. S. Priyadarshini, M. Bajaj, and I. Zaitsev, "Energy feature extraction and visualization of voltage sags using wavelet packet analysis for enhanced power quality monitoring," Scientific Reports, Nature, vol. 15, p. 2226, Jan. 2025.